

Review

Communication between residential aged care facilities and the emergency department: A review of the literature



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ABSTRACT

Background: Western countries have encountered an increase in elderly patients transferred from residential aged care facilities to emergency departments. This patient cohort frequently experiences impaired physical and cognitive function. Emergency department staff require important clinical and personal patient information to provide quality care. International studies show that documentation and handover deficiencies are common.

Objective: The purpose of this literature review was to explore transitional communication practices, and to consider the specific patient information deemed essential for the management of residents in the emergency department.

Methods: A literature review was conducted to examine the studies exploring the documentation accompanying elderly people who were transferred from residential aged care facilities to emergency departments. Scopus, OVID Medline and Cinahl Plus data bases were searched using combinations of the following key words: 'nursing home', 'long-term care', 'skilled nursing facility', 'aged care facility', 'communication', 'documentation', 'emergency department', 'emergency room', 'hospital', 'acute', 'transfer', and 'transition'. Additional data was located with the use of Google Scholar. Review of titles and exclusion of duplicates identified 69 relevant studies. These 69 papers were independently reviewed by three members of the research team for eligibility for inclusion in the review, and seven papers were retained.

Results: There is currently no consensus regarding what information is essential when residents are transferred from aged care facilities to emergency departments, and practices vary. Key information which should accompany the resident has been reported by various authors and include the reason for transfer, past medical history, current medications, cognitive function and advance directives. Some authors also suggest that facility contact details are essential. Without agreement by key stakeholders as to what constitutes 'essential transfer information', clinical practices will continue to vary and resident care will be affected.

Conclusion: This paper identifies frequent communication deficits in the information provided to the emergency department from aged care facilities. There is an imperative to identify suitable items of information which health care professionals agree are essential. Future research should focus on methods to improve the transfer of information between facilities, including consensus regarding what information is essential transfer data.

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What is already known about the topic?

- Older people have higher rates of illness and are over-represented in emergency department presentations.
- Emergency departments may not be the best option for treating people transferred from residential aged care, as this group have a longer length of stay, higher complication rates and in some cases no need of emergency care.
- There is often a communication gap with limited or inadequate patient history information being provided from residential aged care to emergency department.

What this paper adds

- Current research demonstrates that there is no consistency in how communication is managed between residential aged care facilities and emergency departments.
- Of the existing international research there is no agreement as to the key data necessary for the transfer.
- Standardised transfer information forms used in the United States and Canada have shown promise by improving the quality of information, however universal use was not achieved.
- Further research is required, particularly in the areas of patient outcomes, including respecting patient wishes, duplication of investigations or unnecessary interventions, financial cost of the episode of care, and the impact on the emergency department work load.

1. Introduction

Older people have a greater need for emergency medical care than others in the community, and the number presenting to emergency departments from residential aged care is increasing (Edwards et al., 2012; Fealy et al., 2012; Kessler et al., 2013; Roberts et al., 2008). A systematic review examining how frequently older people living in aged care facilities present to emergency departments found that the incidence of transfer is at least 30 transfers per 100 aged care facility beds per year (Arendts and Howard, 2010).

Older patients (aged over 65 years) presenting to emergency departments often present with non-specific symptoms such as falls or confusion, making clinical assessment more complex (Kilshaw, 2009). In the case of elderly patients transferred from aged care facilities, many arrive via an emergency ambulance (Carter et al., 2009; Codde et al., 2010), are unaccompanied by a relative or carer, and are unable to provide a coherent medical history or describe their medication regime (Terrell and Miller, 2007). In addition, the quality of verbal and/or written handover given by paramedic staff (Yong et al., 2008) varies considerably (Owen et al., 2009), because it is based on the information provided by the aged care facility, which may be deficient.

The provision of timely and relevant patient information is vital for assessment and medical management of older patients presenting from aged care facilities (Salinas and Ramakrishnan, 2012). Emergency department staff are reliant on information regarding a patient's clinical,

functional and social circumstances being communicated to them to provide appropriate person-centred care and avoid medical errors (Salinas and Ramakrishnan, 2012). However, international studies have identified a gap in communication between aged care facilities and emergency departments with either limited information, or a deficiency of relevant information being provided (Cwinn et al., 2009; Dalawari et al., 2011; Kessler et al., 2013; Lamantia et al., 2010; Platts-Mills et al., 2012; Terrell et al., 2005). Emergency department staff require important clinical and personal information including the medical history, the reason for transfer to the emergency department, presenting symptoms and any advance directives or patient wishes regarding therapy (Salinas and Ramakrishnan, 2012). When this information is deficient, the quality of care may be negatively affected, with patients at risk of service duplication, or inappropriate and unwanted care (Coleman, 2003). It is therefore important to develop effective methods of communication between services (Cwinn et al., 2009; Terrell and Miller, 2006).

Insufficient communication between residential aged care facilities and emergency departments also impedes treatment and care, contributing to emergency department delays (Cwinn et al., 2009; Finn et al., 2006; Givens et al., 2012). Prolonged length of stay in the emergency department has been shown to result in an increased morbidity and mortality (Australian Medical Association, 2010; Kilcoyne and Dowling, 2008; Richardson et al., 2009; Sprivulis et al., 2006). Moreover, service duplication and delays in emergency care, increase demand on emergency department services and costs to the health care system.

This paper aims to review the current literature to explore transitional communication practices, and to consider the specific information deemed essential for the management of residents in the emergency department.

2. Methods

The literature published in refereed journals between the year 2000 and 2013 was examined. This period was chosen to maintain the currency of evidence, given changes in medicine and in technology over the last decade. Searches were made of the electronic databases Scopus, OVID Medline and Cinahl Plus for peer-reviewed publications in English, and where abstracts were available. Multiple searches were conducted using strategies suitable for each database. For example, the Scopus search strategy began with the following terms: 'nursing home', 'long-term care', 'skilled nursing facility' and 'aged care facility' which were combined using the Boolean search operator "OR". These findings were then combined using "AND" with the following terms: 'communication' "OR" 'documentation', "AND" 'emergency department' "OR" 'emergency room' "OR" 'hospital' "OR" 'acute' "AND" 'transfer' "OR" 'transition'. This search resulted in 7989 results. The search was extended with use of the Google Scholar search engine and an examination of the reference list of located papers, and 32 additional papers were located.

As shown in Fig. 1, duplicate papers were removed, and records were examined for relevance. 4903 papers were excluded based on the title of paper and subject headings, leaving 69 papers of interest. Each abstract was then reviewed by three members of the research team to identify potentially eligible studies applying the following inclusion criteria: quantitative or qualitative designs that identified or examined the documentation accompanying residents from aged care facilities to emergency departments. Fifty-four papers were excluded following abstract review. Some examples of the reasons papers were excluded include: studies which failed to examine the documentation accompanying residents transferred from aged care facilities to emergency departments, studies which examined transfer documentation but in which the resident was not transferred to an emergency department, studies which did not clearly report the outcomes measured (for example, those studies which summarised study findings without presenting them).

Following full-text review of the remaining 15 papers, the three researchers agreed that seven studies be included for the final review. Papers were included if the fundamental research structure was transparent, with the method and study design clearly described and outcomes reported. Owing to the various study designs,

the studies were reported using a narrative framework and with a focus on communication practices.

3. Results

Seven studies were located which examined the documentation accompanying residents transferred from aged care facilities to emergency departments (Boockvar and Fridman, 2006; Cwinn et al., 2009; Dalawari et al., 2011; Davis et al., 2005; Platts-Mills et al., 2012; Terrell et al., 2005; Zafirau et al., 2012). Most authors undertook a retrospective review of the transfer documentation in resident medical histories (Boockvar and Fridman, 2006; Cwinn et al., 2009; Dalawari et al., 2011; Terrell et al., 2005; Zafirau et al., 2012). One study surveyed physicians about the presence or absence of transfer documentation, immediately after they had cared for a resident transferred from an aged care facility (Platts-Mills et al., 2012), while another study surveyed emergency staff about their perceptions of changes required to the quantity and quality of transfer documentation, following the introduction of a standardised transfer form (Davis et al., 2005). The number of histories audited in these studies ranged from 80 (Dalawari et al., 2011) to 457 (Cwinn et al., 2009). A summary of these papers is provided in Fig. 2 and each of the studies are reviewed in the following sections.

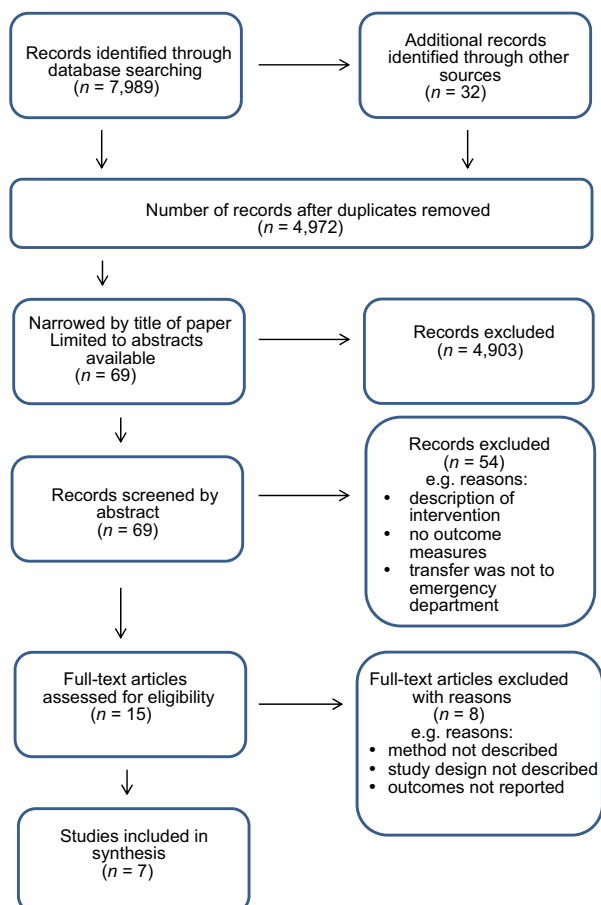


Fig. 1. PRISMA strategy used to identify literature.

3.1. Characteristics of residential aged care facility residents presenting to emergency departments

Individuals transferred from residential aged care facilities to emergency departments are principally the elderly who are in their seventh, eighth or ninth decade (Boockvar and Fridman, 2006; Cwinn et al., 2009; Dalawari et al., 2011; Platts-Mills et al., 2012; Zafirau et al., 2012). Residents were referred to the emergency department by staff in the aged care facility (Cwinn et al., 2009), and arrived via ambulance. The primary reasons for transfer to the emergency department were: respiratory tract infections (Boockvar and Fridman, 2006; Cwinn et al., 2009; Zafirau et al., 2012), trauma including falls and fractures (Cwinn et al., 2009), neurological disturbances (Cwinn et al., 2009), and urinary tract infections (Boockvar and Fridman, 2006).

3.2. Documentation and handover requirements for residents presenting to the emergency department

Several studies have identified key information or data which should accompany residents transferred from aged care facilities to emergency departments (Boockvar and Fridman, 2006; Cwinn et al., 2009; Dalawari et al., 2011; Davis et al., 2005; Platts-Mills et al., 2012; Terrell et al., 2005). Fig. 3 provides a summary of the key information identified by the seven studies included in this review as being essential to the transfer process. As shown in Fig. 3, the most common include: reason for transfer ($n=6$); baseline cognitive function ($n=6$); advance directives ($n=5$); current medications ($n=5$); and past medical history ($n=4$). There is commonality in this most frequently cited data. However, there is considerable

Author	Country	Method	Study design	Outcome measures	Sample size	Results
Boockvar & Fridman (2005)	USA	Retrospective medical history audit	Evaluation post education regarding privacy principles	12 essential criteria required for transfer	100 transfers	86.7% of documents were complete. Less complete transfers were associated with older females with dementia, shorter duration in aged care facilities & transfers occurring after hours
Cwinn et al. (2009)	Canada	Retrospective medical history review of all resident transfers in 6 month period	Comparison of data documented with & without a standardized transfer form	11 essential criteria required for transfer	457 transfers	Important information gaps occurred in 85.5% of cases
Dalawari et al. (2011)	USA	Retrospective medical history review of all resident transfers in 6 month period	Comparison of data documented with & without a standardized transfer form	16 essential criteria were identified and weighted to reflect importance in Emergency department patient care	80 transfers	Residents transferred with a transfer form had 71% of essential items present compared to 28% for cases without a transfer form ($p < 0.001$). Following education in aged care facilities use of the form increased from 30% to 90%
Davis et al. (2005)	USA	Mixed method 1. Two ED staff surveys 2. Quantitative evaluation of standardized transfer form use.	Developed a standardized transfer form & evaluated staff perceptions re the effectiveness of the form	1. Staff perceptions of transfer information 2. Quantitative analysis of rate of form use.	Survey 1 - 120 ED nurses No sample size reported for survey 2 or evaluation	
Platts-Mills (2012)	USA	Survey of ED physicians post patient examination	Physicians identified the information accompanying patients from Medicaid and Non-Medicaid funded aged care facilities	9 essential criteria were identified based on previous studies	128 patient assessments	Less information was available from Medicaid funded aged care facilities than non-Medicaid funded facilities (7.13 vs 8.15, $p < 0.001$)
Terrell et al. (2005)	USA	Retrospective medical history review pre and post intervention	Comparison of data documented pre & post introduction of a 1 page standardized transfer form	11 criteria required for transfer were identified based on previous studies. At least 9 criteria necessary for successful documentation transfer	130 transfers pre and 72 transfers post intervention	Post intervention documentation increased from 58.5% to 77.8%
Zafirau et al. (2012)	USA	Retrospective medical history review pre and post intervention	Comparison of data documented pre & post introduction of a 1 page standardized transfer form	Documentation of advance directives	130 pre and 117 post intervention	Post intervention concordance in documented advance directives increased by 12.4% ($p = 0.038$).

Fig. 2. Studies examining documentation accompanying patients transferred from aged care facilities to emergency departments (ED).

variability in the remaining data identified in the literature with multiple items identified as necessary. Moreover, some authors have prioritised the data, identifying some data as essential (Cwinn et al., 2009; Dalawari et al., 2011).

Ideally, documented patient information should be available to emergency department staff at the time of the patient's arrival. This should be in concise and organised formats that are easily read (Terrell and Miller, 2006).

Essential data that should accompany each patient transferred	Terrell et al. (2005)	Davis et al. (2005)	Boockvar and Fridman (2006)	Cwinn et al. (2009)	Dalawari et al. (2011)	Platts-Mills et al. (2012)
Reason for transfer to emergency department	✓	✓	✓	✓	✓	✓
Baseline cognitive function (mental status) and communication ability	✓	✓	✓	✓	✓	✓
Advance directives for level of care and resuscitation	✓	✓		✓	✓	✓
Current medications		✓	✓	✓	✓	✓
Past medical history		✓	✓		✓	✓
Vital signs at time of complaint			✓	✓	✓	
Activities of daily living/ Usual functional status	✓		✓	✓		
Healthcare power of attorney / Patient proxy			✓		✓	✓
Allergy status				✓	✓	✓
Mobility/ baseline ambulatory status			✓	✓		✓
Primary care physician	✓		✓		✓	
Aged care facility charge nurse	✓		✓		✓	
Aged care facility phone number	✓	✓			✓	
Patient's name	✓				✓	
Patient's date of birth	✓				✓	
Aged care facility name	✓					✓
Bladder continence			✓	✓		
Bowel continence			✓	✓		
Laboratory data			✓		✓	
Immunization status if wounded				✓		
Family member name and contact number		✓				
Patient's Social Security number	✓					
Capabilities of aged care facility					✓	
Wing or room of resident					✓	

Fig. 3. Essential transfer information.

However, several studies have reported that information gaps are prevalent when residents are transferred to the emergency department. In Canada, an audit of transfer documentation accompanying residents from aged care facilities found information gaps in 93.5% of 457 transfers (Cwinn et al., 2009). Important information which was commonly absent included baseline cognitive function and baseline orientation information. Further, 7.7% of residents who were reported to be experiencing dementia, had no reason for transfer specified (Cwinn et al., 2009), making decisions about investigations and management difficult. Similar findings were reported in a USA records audit, which identified that 41.5% of 65 transfers from extended care facilities had inadequate documentation (Terrell et al., 2005).

Conversely, in the United States an audit of medical records for individuals transferred between a nursing home and a hospital examined 12 items that were regarded as necessary information for continuity of the patient's care (Boockvar and Fridman, 2006) (Fig. 2). Of 100 records, 86.7% were complete. Interestingly, this study was undertaken to explore if there was a reduction in the information accompanying residents transferred from aged care facilities to emergency departments following the introduction of, and education about, new privacy rules. The authors found no change in the documentation accompanying residents after the introduction of the privacy rules (Boockvar and Fridman, 2006).

In addition, United States emergency physicians audited the documentation of nine essential information items for transferred 'skilled nursing facility' patients ($n = 128$) aged ≥ 65 years, and found that on average, 7.6 of nine items were documented (Platts-Mills et al., 2012) (Fig. 2). Of concern, the information which was commonly absent were medication charts and advance directive status (Platts-Mills et al., 2012), information which has been reported to be essential by other authors (Cwinn et al., 2009; Dalawari et al., 2011; Terrell et al., 2005).

Documentation and handover deficiencies have thus been identified across a number of countries. Characteristics associated with incomplete transfer documentation include: older age; female gender (Boockvar and Fridman, 2006; Platts-Mills et al., 2012); a dementia diagnosis; and a shorter duration of nursing home residence (Boockvar and Fridman, 2006). Studies have also shown that there are more gaps in documentation when the transfer occurs outside office hours (Boockvar and Fridman, 2006; Dalawari et al., 2011).

3.3. Effectiveness of standardised patient transfer forms

Recommendations that have been made for improving patient safety during medical treatment include the use of more structured and explicitly designed forms of communication to reduce ambiguity, enhance clarity, and signal that specific action is required (Dayton and Henriksen, 2007). Terrell et al. (2005), Zafirau et al. (2012), and Davis et al. (2005), undertook interventional studies, introducing transfer forms to local aged care facilities. Terrell et al. (2005) and Zafirau et al. (2012) examined the documentation that accompanied residents before and after the

introduction of the transfer form. Terrell et al. (2005) reported that the amount of essential data transferred between aged care facilities and the emergency department increased following the introduction of the transfer form. Further studies are necessary to confirm that improvements in transfer documentation are sustained.

Similarly, Zafirau et al. (2012) recently reported improved communication of advance directives between aged care facilities and acute care, through the introduction of a transfer form for 26 long-term care facilities in southern United States (Fig. 2). In this study, a form comprising patient demographic and medical data was developed and implemented following staff training. The transfer forms were only found to be used for half of the residents transferred to emergency departments ($n = 72$, 50%), however, an audit of documentation (130 before; 117 after) showed significant improvement in the concordance between documented advance directives in the aged care facility and the emergency department, increasing from 66.6% to 79% ($p = .038$) following the introduction of the form. As such, this form was found useful for resident transitions both to, and from acute care, and it was considered to be comprehensive, yet easy to complete.

Davis et al. (2005) developed their transfer form based on the information emergency staff reported they wanted to know. The transfer form was introduced in local aged care facilities, and education was provided to aged care facility staff. Davis et al. (2005) surveyed the emergency department staff following the introduction of the transfer form, and reported that emergency staff perceived there had been an improvement in the communication between aged care facilities and the emergency department. The findings in this study could be strengthened by a review of the documentation accompanying residents, to substantiate the reported perceptions.

In contrast to the previous three studies, Cwinn et al. (2009) and Dalawari et al. (2011) did not introduce or promote a standardised transfer form, instead they examined the use of a transfer form which had been in place for many years. The authors compared the information which accompanied a resident when transferred with the form, to the information accompanying a resident when they arrived to the emergency department without the transfer form. Both studies reported that more essential information accompanied the resident who was transferred with the standardised transfer form, than the resident who was transferred with an institutional form, or no form at all (Cwinn et al., 2009; Dalawari et al., 2011).

4. Discussion

In order to provide quality care to residents, emergency department staff require accurate and complete health information (Salinas and Ramakrishnan, 2012). Various authors identify key information which ideally should accompany each patient (Boockvar and Fridman, 2006; Cwinn et al., 2009; Dalawari et al., 2011; Davis et al., 2005; Platts-Mills et al., 2012; Terrell et al., 2005). A literature review by Terrell and Miller (2007) examining the epidemiology of residential transfers and how information is communicated, identified that communication deficits

between aged care facilities and emergency departments were common. However, beyond advance directives, the authors did not examine which types of information accompanied residents, nor what information was deemed important (Terrell and Miller, 2007).

There is evidence that patients frequently arrive to emergency departments without necessary information. It is well established that reason for transfer, baseline cognitive function, advance directives, and current medications are essential transfer data (Boockvar and Fridman, 2006; Cwinn et al., 2009; Dalawari et al., 2011; Davis et al., 2005; Platts-Mills et al., 2012; Terrell et al., 2005). Despite the emphasis on the use of standardised transfer forms (Cwinn et al., 2009; Dalawari et al., 2011; Davis et al., 2005; Terrell et al., 2005), there is disparity in the information requested within these forms. For example, some authors focus on administrative information such as social security details and the contact information of the aged care facility and charge nurse (Terrell et al., 2005). The absence of consistent guidelines makes it difficult for aged care facility staff to provide the requisite information. Further, the absence of adequate information makes it difficult for emergency staff to make informed decisions, which may result in excessive or unwanted investigations and inefficient patient management that is not patient-centred. Unnecessary treatment and investigations contribute to patient and family distress, increased length of stay in emergency departments and increased health care costs, in a currently unsustainable health system (Finucane et al., 1999).

Consensus regarding what constitutes essential transfer information must be reached, to facilitate the provision of patient-centred care in emergency departments. Key information which the majority of authors agree is essential includes: reason for transfer; baseline cognitive function; advance directive; current medications; and past medical history. Agreement on these and other key information needed, is necessary to enable a consistent message to be delivered to aged care staff.

Although the use of paper-based standardised transfer forms improved transfer of information (Cwinn et al., 2009; Dalawari et al., 2011; Terrell et al., 2005; Zafrau et al., 2012), application is dependent upon education and training of the aged care facility staff who will be responsible for its completion. Few details of the extent of staff education were provided and no details of policy development were evident. There are indications that the use of standardised transfer forms declines with time following the initial introduction of, and education about, the transfer forms (Cwinn et al., 2009; Dalawari et al., 2011). Regular reminders of the importance of standardised transfer forms may address this issue. In addition, various other factors relating to aged care facilities form a barrier to the use of standardised transfer forms including: low staff ratios, funding for education, presence of less qualified staff, transfer policies, and involvement of a General Practitioner. Thus, there is no universal panacea to the communication problem between aged care facilities and emergency departments.

Further efforts to improve communication have involved the use of the Internet for communicating between

aged care facilities and emergency departments. The introduction of electronic health records in unrelated organisations have shown this communication is feasible (Brandeis et al., 2007). A United States study of 276 emergency department transfers using a trial communication network, identified barriers to universal use of this system (Hustey and Palmer, 2012). The system was used in 28% of transfers, with barriers including a lack of electronic medical record use at aged care facilities, staff time required for system use, and negative staff attitudes. No examination of the quality or quantity of information transferred using this system was undertaken. While feasible, such communication requires optimum electronic communication networks and policy and practice commitment.

The literature demonstrates a focus on the communication between aged care facilities and emergency departments, with a particular emphasis on standardised patient transfer forms. The quality of each study is affected by the collection of data from single emergency departments. Sample size was also a limitation, in that several studies used small samples (<150) (Boockvar and Fridman, 2006; Dalawari et al., 2011; Platts-Mills et al., 2012; Terrell et al., 2005; Zafrau et al., 2012) or did not report sample size (Davis et al., 2005). Despite these limitations, the common findings add weight to each author's conclusions.

Importantly, there was little consensus regarding what information was essential in the transfer of residents from aged care facilities to the emergency department. Three studies relied on consensus among emergency department staff to identify information which was essential, and this may have resulted in site specific bias (Dalawari et al., 2011; Davis et al., 2005; Terrell et al., 2005). To address this potential bias, Dalawari et al. (2011) also considered the essential transfer information identified in the literature. The remaining three studies relied solely on the literature to identify important transfer information (Boockvar and Fridman, 2006; Cwinn et al., 2009; Platts-Mills et al., 2012). Further areas for future research include the effect of standardised transfer forms on patient outcomes, including respecting patient wishes, advance care planning, duplication of investigations or unnecessary interventions, financial cost of the episode of care, and the impact on emergency department work load.

5. Conclusion

An increasing number of residents from aged care facilities are being transferred to emergency departments for management. This cohort represent the sickest and most vulnerable members of our community, who are often unable to advocate for themselves, due to illness, confusion or deterioration. The provision of quality emergency care to this group of patients is dependent on clear, concise and contemporary communication regarding their relevant medical history, current medications, reason for transfer, baseline cognition and advance directives. This paper identifies frequent communication deficits in the information provided to the emergency department from aged care facilities. Such deficits may result in unnecessary investigations and interventions.

There is an imperative to identify suitable items of information which health care professionals agree are essential. Standardised transfer forms have shown limited improvements in the quality of communication between aged care facilities and emergency departments, and this is not unexpected given the diversity of information identified as essential. Future research must focus on improving the transfer of information between aged care facilities and emergency departments; to ensure the care provided best meets the patient's and family's needs and wishes.

Conflict of interest

None declared.

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